COMP 8790 · TOPICS IN APPLIED AI · PROJECT PROPOSAL

Does timing matter? Honest AI for spotting mental-health risk online

A 2025 paper says that knowing *when* someone posts helps an AI spot depression, anxiety and suicidality on Reddit. The gain it reports is tiny, it has never been checked against a stronger text model, and it was never tested on posts from a later time period. We test it.

Domain: Healthcare / Mental health | Task: NLP text classification | Data: public, 700k+ posts | Compute: Colab T4

THE HOOK

WHAT WE FOUND READING THE ANCHOR PAPER

The +1.5 F1 "temporal" gain is small, untested for significance, and the method isn't really early detection



Reported F1 from Saeed & Ahmed (2025), Table III. Bars start at zero.

- The "temporal" features are just the hour, weekday and month of **each individual post**. There are no user histories, so it isn't early detection in the eRisk sense.
- The results are a single **validation** split, with no seeds, confidence intervals or released code.
- There is **no transformer baseline** such as MentalBERT, which is the standard for this task since 2022.
- There is **no robustness test**: nobody knows whether it still works on posts from a later year.

The course asks, "What did we learn that was not obvious from reading the papers?" This paper gives us that question on day one.

FOR THE TEAM

Why we should pick this one

1. **The research finding is built in.**
   Phase I is graded on an evidence-based finding. We already have a testable claim to confirm or overturn: either the timing features help or they don't. Both outcomes are publishable findings.

2. **Reproduction fits the rules exactly.**
   The course says running the authors' code is not enough. The anchor paper has no code, so we have to reimplement it, and the paper gives full equations and hyperparameters (Sections V–VI).

3. **The data is free, large and downloaded today.**
   The paper's own Kaggle set has about 700k posts with timestamps. The Low et al. Zenodo set is public domain, 3.1 GB, and has pre- and mid-COVID splits. Neither needs a data agreement.

4. **It runs on Colab.**
   TF-IDF + LR takes minutes, the BiLSTM runs on a laptop GPU, and MentalRoBERTa-base fine-tunes in about 1–2 hours on a T4. There are no API costs.

5. **Phase II follows directly.**
   Our Phase I results decide how the Phase II system is designed. Whichever model is most reliable over time becomes the core of a triage assistant for volunteer moderators, with explanations and a human always in the loop.

6. **It matters.**
   Suicide kills about 727,000 people a year and is a leading cause of death for 15–29-year-olds. The people who post in these communities are often not seen by anyone else.

7. **There's a shot at the +5% bonus.**
   A careful reproduction plus a robustness study is a recognised paper type at workshops such as CLPsych.

READY TO PASTE · WEEK 2 FORM (5%)

Problem Selection & Feasibility

Header

Group Number: [fill in]
Selected Domain: Healthcare (Mental Health)
Project Title: Does Timing Matter? Evaluating Temporal Signals and Time-Robustness in AI Detection of Mental-Health Risk from Reddit Posts

1. Problem Definition

Many people describe their distress in online communities such as Reddit long before, or instead of, seeing a clinician. Recent research proposes AI models that read these posts and flag signs of depression, anxiety, bipolar disorder, borderline personality disorder and suicidality. A 2025 study (Saeed & Ahmed, arXiv:2503.07653) claims that adding posting-time features (hour, weekday, month) to a text model through cross-modal attention improves detection. However, the reported gain is small (F1 0.7376 vs. 0.7222 for a text-only BiLSTM), comes from a single validation split with no released code, and was never compared against domain-adapted language models such as MentalBERT. It is also unknown whether these models stay reliable as the language of distress changes over time, for example between pre- and mid-COVID-19 posts.

We will investigate the specific problem of classifying the mental-health risk category of Reddit posts. We will test (1) whether temporal metadata adds real predictive value beyond text, and (2) how robust text-only and text+temporal models are under temporal distribution shift. We will reimplement three representative methods (TF-IDF + logistic regression, fine-tuned MentalRoBERTa, and BiLSTM with cross-modal attention) and compare them under identical, leakage-free splits on public datasets (the Kaggle Mental Disorders Reddit dataset, about 700k posts, and the Low et al. 2020 Zenodo dataset).

2. Motivation and Real-World Relevance

Why it matters: Around 280 million people live with depression and 301 million with anxiety disorders (WHO), and suicide caused about 727,000 deaths in 2021. Suicide is a leading cause of death among 15–29-year-olds. In low- and middle-income countries, more than 75% of people with mental disorders receive no treatment, and people often wait years between the first symptoms and first care.

Who is affected: People in distress who post in online support communities, especially young people (nearly half of US teens say they are online "almost constantly", Pew 2024). Also the volunteer moderators and peer supporters who must read huge volumes of posts, and the counselling and crisis services that could reach people sooner.

Practical benefit: An AI triage tool could bring high-risk posts to trained human moderators faster, with calibrated confidence and an explanation of which words drove the flag. It would support human judgement, not replace it or diagnose anyone. Just as important, our evaluation shows which signals can be trusted. It checks whether a claimed improvement is real and whether a model degrades over time, so that communities don't deploy a tool that quietly fails the people it is meant to protect.

3. Proposed AI Task

Supervised multi-class text classification (NLP).
Input: a Reddit post (title + body) and its posting-time metadata (hour, weekday, month).
Output: the predicted mental-health category (e.g. depression, anxiety, bipolar, BPD, schizophrenia, suicidality, or a control/non-clinical class), with a calibrated probability and a word-level explanation.

We will reimplement and compare three methods under identical splits and metrics (macro-F1, recall on high-risk classes, AUPRC, calibration error):
(A) TF-IDF + logistic regression (classical baseline);
(B) fine-tuned MentalRoBERTa (domain-adapted transformer);
(C) BiLSTM + LSTM with cross-modal attention over text and temporal features (Saeed & Ahmed, 2025).

Research question: Do temporal features significantly improve detection over a strong text-only model, and how much does each approach degrade when tested on posts from a later time period than it was trained on?

Phase II: an end-to-end, human-in-the-loop triage assistant for community moderators. It ranks incoming posts by risk, explains each flag, and shows crisis resources.

THE GOAL

What "done" looks like

By Nov 6	Reimplement methods A, B and C on one shared, leakage-free split. Reproduce the anchor's F1 of 0.7376 to within ±3 points, or explain the gap.
Finding 1	Measure how much the timing features add, as an ablation over 5 seeds with bootstrap 95% confidence intervals. The target is a sentence like: *"Timing adds X ± Y F1 to a BiLSTM but Z to MentalRoBERTa."*
Finding 2	Measure the macro-F1 drop when training on earlier posts and testing on later ones (2021→2022, and pre-COVID→mid-COVID), per model family.
By Dec 4	Deliver a working triage assistant that beats the TF-IDF baseline on macro-F1 and on high-risk recall, with calibration (ECE) and failure cases reported.
Stretch	With CLEF eRisk access, extend to user-level early detection scored by ERDE. Aim for a publication-ready report (+5%).

WHY IT MATTERS

The numbers

280M	301M	727k	>75%
people living with depression (WHO)	living with anxiety disorders (WHO)	suicide deaths in 2021 (WHO)	untreated in low- and middle-income countries (WHO)

Sources: WHO depression fact sheet, WHO mental disorders fact sheet, Pew 2024. Check the latest WHO wording before submitting.

LITERATURE · 5 CORE + EXTENDED

Papers

PAPER	YEAR	ROLE	METHOD	CODE / DATA
Saeed & Ahmed: Early Detection of Mental Health Issues Using Social Media Posts	2025	Core Reproduce C	BiLSTM + LSTM + cross-modal attention	No code · Kaggle data
Ji et al.: MentalBERT	2022	Core Reproduce B	Domain-adapted BERT/RoBERTa	HF weights (gated)
Low et al.: NLP of Reddit Mental Health Communities During COVID-19 (JMIR)	2020	Core Reproduce A	TF-IDF + linguistic features, LR/SGD	Code · Zenodo
Parapar et al.: Overview of eRisk 2025	2025	Core	Early-risk benchmark, ERDE / F-latency	Data under agreement
Cao et al.: Systematic review of biases (47 studies)	2024	Core	Methodology critique	n/a
DS@GT at eRisk 2025	2025	Extended	MentalRoBERTa + temporal attention	See paper
SINAI at eRisk 2025	2025	Extended	Transformers, conversational	Unverified

PAPER	YEAR	ROLE	METHOD	CODE / DATA
Social Media for Mental Health: Data, Methods, Findings (survey)	2025	Extended	Survey	n/a
Menta: on-device small LM for mental health	2025	Extended · Phase II	Distilled small LM	See paper
Mental Disorder Classification via Temporal Representation of Text	2024	Extended	Temporal compression of post history	Unverified
Yates et al.: RSDD · Cohan et al.: SMHD	2017–18	Background	User-level benchmarks	Data under agreement

Two 2026 eRisk papers (arXiv 2609.08161, 2605.14995) came up in search, but we haven't opened them yet. Read them before citing.

DATA

Datasets

DATASET	SIZE	TIMESTAMPS	ACCESS	USE
Kaggle: Mental Disorders Identification (Reddit)	~701k posts	Full UTC	Open (Kaggle login)	Anchor reproduction, timing ablation, 2021→2022 shift
Low et al. Reddit Mental Health (Zenodo)	3.1 GB, 28 subreddits	Day only	Public domain	Pre- vs mid-COVID shift, control subreddits
Kaggle: Suicide and Depression Detection	232k posts	No	Open	Cross-dataset test

DATASET	SIZE	TIMESTAMPS	ACCESS	USE
CLEF eRisk	~909 users (2025)	User timelines	Agreement	Stretch goal: true early detection

What we checked in the Zenodo files: each file has `subreddit, author, date, post` plus 346 precomputed features. Most authors appear only once per period (74 of 782 bipolar users recur between 2018 and 2019), so we evaluate at the post level.

PLAN

Timeline against course deadlines

- **Sep 25** — Submit this form. Post the domain on the Brightspace forum first. Apply for eRisk access the same day.
- **Oct 2** — Literature comparison matrix of 10–12 papers and a final research question.
- **Oct 9** — Proposal: the shared split protocol (split by time, deduplicated) and the metrics.
- **Oct 23** — Reimplement A, B and C. Compare reproduced and reported results.
- **Oct 30** — Timing ablation and time-shift experiment, over 5 seeds with confidence intervals.
- **Nov 6** — Phase I presentation and conference-style report.
- **Nov 13–27** — Triage assistant: model service, explanation layer, moderator interface, evaluation.
- **Dec 4** — Demo, technical report, responsible-AI analysis.

FIVE PEOPLE, FIVE CLEAR ROLES

Who does what

Data & splits
Loading, deduplication, time-based and leakage-free splits, dataset documentation

Classical baseline
TF-IDF + LR, the statistics toolkit (bootstrap, McNemar)

Transformer
Fine-tuning MentalRoBERTa, calibration

Anchor reimplementation
BiLSTM + cross-modal attention, timing ablation

System & ethics
Phase II interface, explanations, responsible-AI report

Everyone reviews the literature and the final report together, because the course requires every member to explain the whole project.

HONEST RISKS

What could go wrong, and our plan

Labels come from the subreddit, not a diagnosis	We frame the task as "risk signal", never diagnosis, and add a cross-dataset test.
Leakage and duplicate posts inflate scores	Deduplicate, split by time, and strip subreddit names from post text.
We can't match the 74.55% reported accuracy	That is still a valid reproduction finding. We report and explain the discrepancy.
Privacy and ethics of screening people	Public, anonymised data only. No quoting of posts. A human decides every flag, and crisis resources are always shown.
eRisk access doesn't arrive in time	It's only a stretch goal. The core plan runs entirely on open data.
The healthcare domain is already taken on Brightspace	Check the forum today. If it's taken, the same project can be framed under a "social media / online safety" domain.